

Lesson 3: Defining Probability

TB sections 2.1

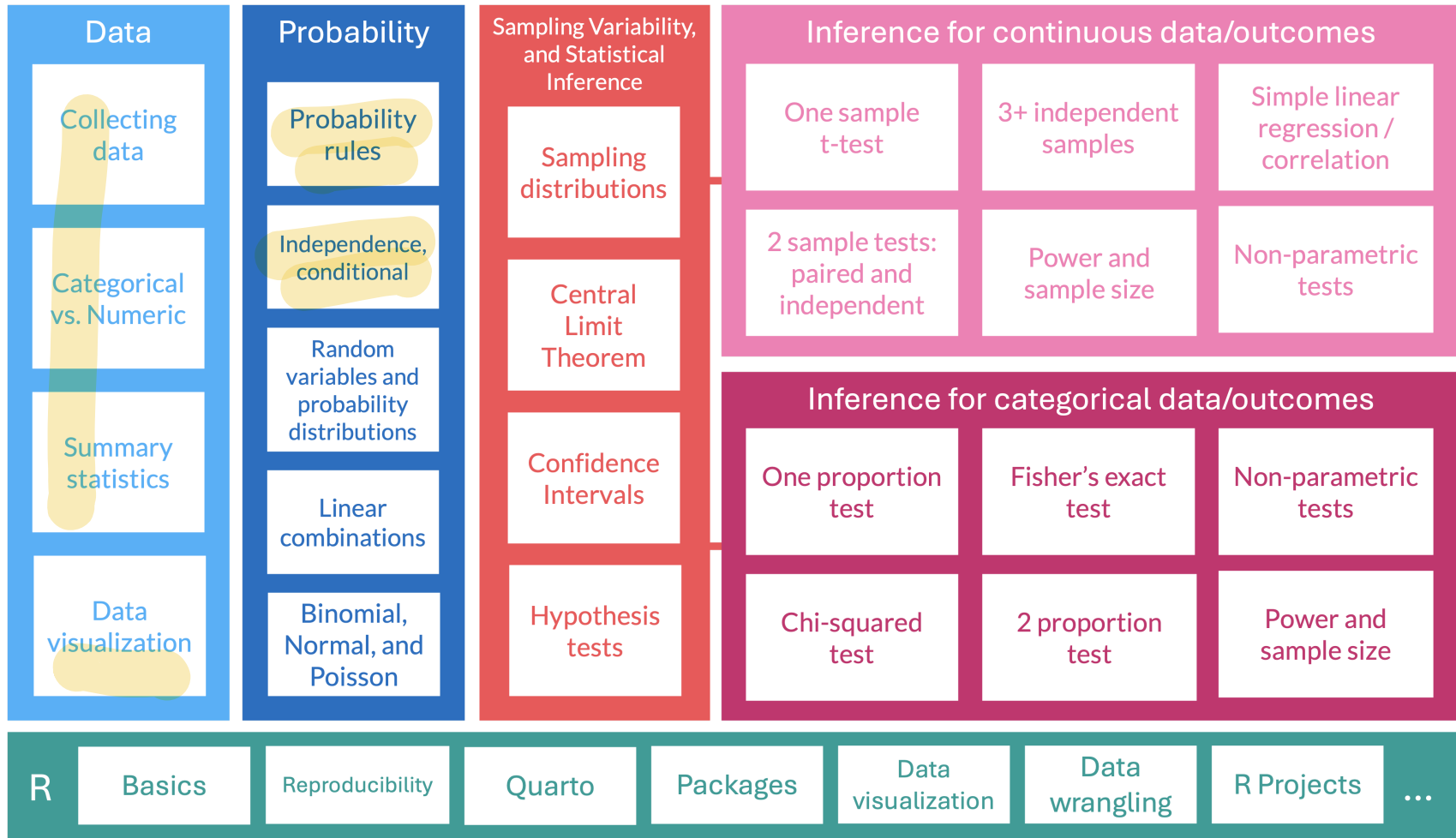
Nicky Wakim

2025-10-06

Learning Objectives

1. Define probability and explain the Law of Large Numbers within examples
2. Define relationships between events and their probability properties (including disjoint events, non-disjoint events, complements, and independent events)
3. Calculate an unknown probability in a word problem using the probability properties

Where are we?



Let's start with an example!

Example: Rolling fair 6-sided dice

Suppose you roll a fair 6-sided die.

1. What is the probability that you roll a 4?
2. What is the probability that you roll an even number?
3. What is the probability that you did not roll a 3?

① 6 possible side
1 side is 4

$$P(\text{getting a 4}) = \frac{1}{6}$$

event

possible
outcomes
w/in event

possible
outcomes
total

② 2, 4, 6 → 3 sides even
 $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$

③ 1, 2, 4, 5, 6

$$P(\text{not a 3}) = \frac{5}{6}$$

Learning Objectives

1. Define probability and explain the Law of Large Numbers within examples
2. Define relationships between events and their probability properties (including disjoint events, non-disjoint events, complements, and independent events)
3. Calculate an unknown probability in a word problem using the probability properties

What is a probability?

Definition: Probability

How likely something will happen.

- On a more technical note, the probability of an outcome is the proportion of times the outcome would occur if the random phenomenon could be observed an infinite number of times.
- We can think of flipping a coin. There are two possible outcomes (heads or tails). The probability of getting heads is 0.5.

What is a probability? with the Law of Large Numbers

- We can think of flipping a coin. There are two possible outcomes (heads or tails). The probability of getting heads is 0.5.
 - If we flip the coin 10 times, it is not certain that we will get 5 heads. However, if we flip it infinite times, we will get heads 50% of the flips.
- Fun “Seeing Theory” demonstration!

10 flips
6 heads

$$\hat{p} = \frac{6}{10} = 0.6$$

Law of large numbers

As more observations are collected, the proportion of occurrences, $\hat{p}$, with a particular outcome converges to the true probability p of that outcome.

$p = 0.5$

Poll Everywhere Question 1

13:24 Mon Oct 6

Join by Web PollEv.com/nickywakim275

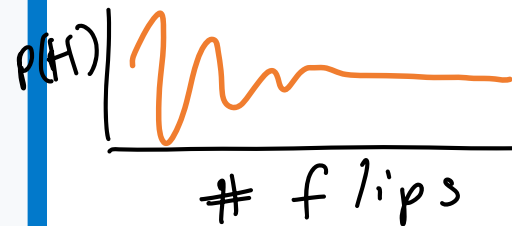


From the coin flip example, which of the following statements is false as they relate back to populations and samples?

Statement	Percentage
The true underlying (population) probability of heads is 0.5	13%
A sample of 100 coin flips is more likely to have exactly 50% heads than a sample of 10 flips	37%
As we continue to sample more and more coin flips, the proportion of heads gets closer to 0.5	0%
It is possible to have 0 heads in a sample of 100 coin flips	50%

Powered by  Poll Everywhere

from 100 from 10
 $P(50 H) \leq P(5 H)$
better resolution
w/ 100 flips



Some probability notation

- Probability typically defined as a proportion
 - Takes values between 0 and 1

$\hat{p}$
sample
prob/
proportion

p
true, pop
prob

$$0 \leq \hat{p} \leq 1 \quad \& \quad 0 \leq p \leq 1$$

- Probability can also be expressed as a “percent chance,” taking values between 0% and 100%

- If we want to discuss the probability of an event, say A, we would write $P(A)$
 - We can write: $A = \{\text{rolling a 1}\}$, with associated probability $P(A) = \frac{1}{6}$
 - OR we can write $P(\text{rolling a 1})$

Learning Objectives

1. Define probability and explain the Law of Large Numbers within examples
2. Define relationships between events and their probability properties (including disjoint events, non-disjoint events, complements, and independent events)
3. Calculate an unknown probability in a word problem using the probability properties

Disjoint / mutually exclusive events

Disjoint / mutually exclusive events

Two events or outcomes are called **disjoint** or **mutually exclusive** if they cannot both happen at the same time.

ONE
coin
flip

event A is getting a head

event B is getting a tail

} cannot
get H
& T in
one
flip

roll
2
dice

event C: sum greater than 8

event D: one of my dice rolls
a 1

Poll Everywhere Question 2

13:38 Mon Oct 6

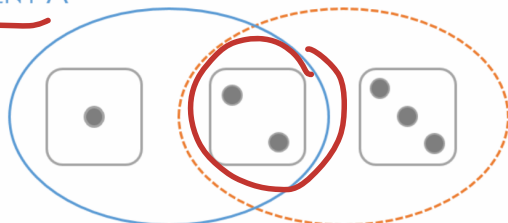


Join by Web PollEv.com/nickywakim275



Let's say we roll a die with events A, B, and D: A is the event of rolling a 1 or 2, B is the event of rolling a 4 or 6, and D is the event of rolling a 2 or 3. Which events are disjoint?

EVENT A



EVENT D

EVENT B

A and B 8%

A and D 8%

B and D 4%

Both (A and B) and (A and D) 0%

Both (A and B) and (B and D) ✓ 76%

Both (A and D) and (B and D) 4%

All events are disjoint 0%

Probability for disjoint events

A, B, C, D name events
 A_1, A_2, A_3, A_4 name events

Probability rule for disjoint events

If A_1 and A_2 represent two disjoint outcomes, then the probability that either one of them occurs is given by

$$P(A_1 \text{ or } A_2) = P(A_1) + P(A_2)$$

If there are k disjoint outcomes $A_1, \dots, A_k$, then the probability that either one of these outcomes will occur is

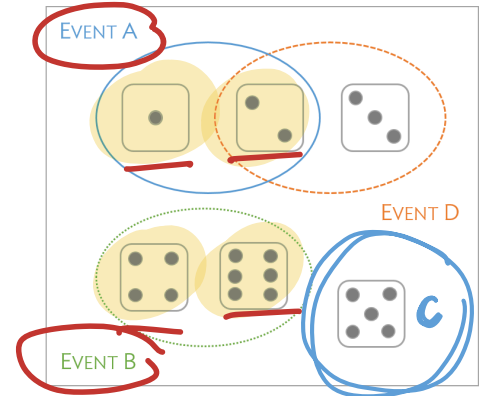
$$P(A_1) + P(A_2) + \dots + P(A_k)$$

$$P(A_1) + P(A_2) + \dots + P(A_{20})$$

- From the poll everywhere question with the die, what is the probability of event A or B?

$$\begin{aligned} P(A \text{ or } B) &= P(A) + P(B) \\ &= \frac{2}{6} + \frac{2}{6} \\ &= \frac{4}{6} = \frac{2}{3} \end{aligned}$$

$$\begin{aligned} P(A \text{ or } B \text{ or } C) &= P(A) + \\ &\quad P(B) \\ &\quad + P(C) \end{aligned}$$



Probabilities when events are not disjoint

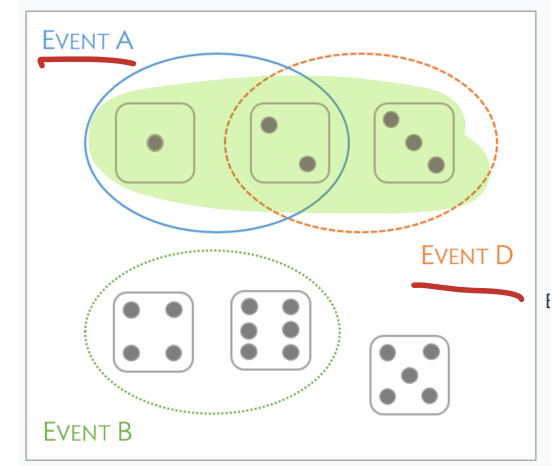
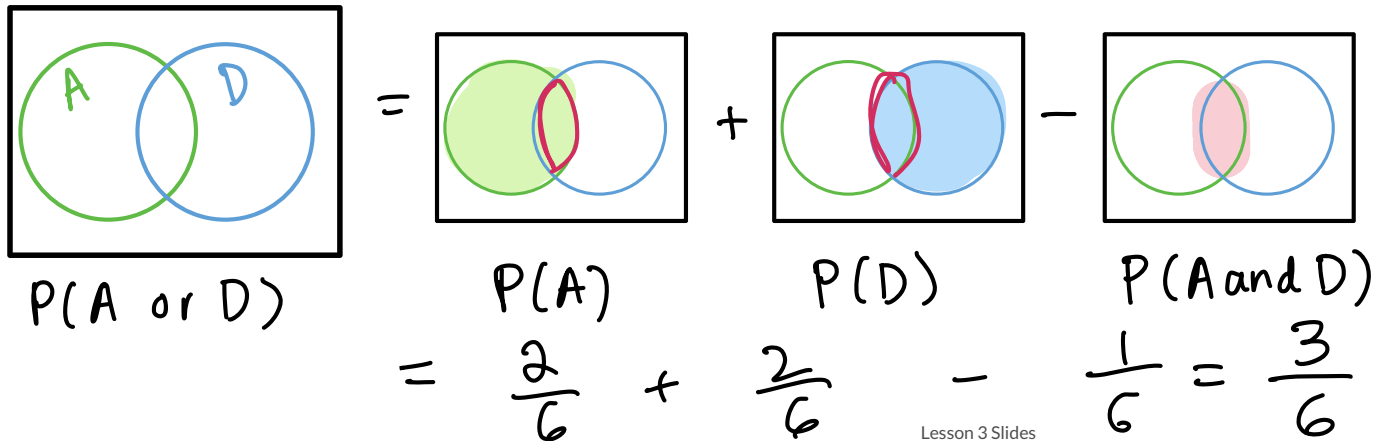
- When events are not disjoint, we cannot use the previous addition rule for probabilities!!
- We must use a general rule that recognizes the potential overlap between events

General probability addition rule

If A and D are any two events, disjoint or not, then the probability that at least one of them will occur is

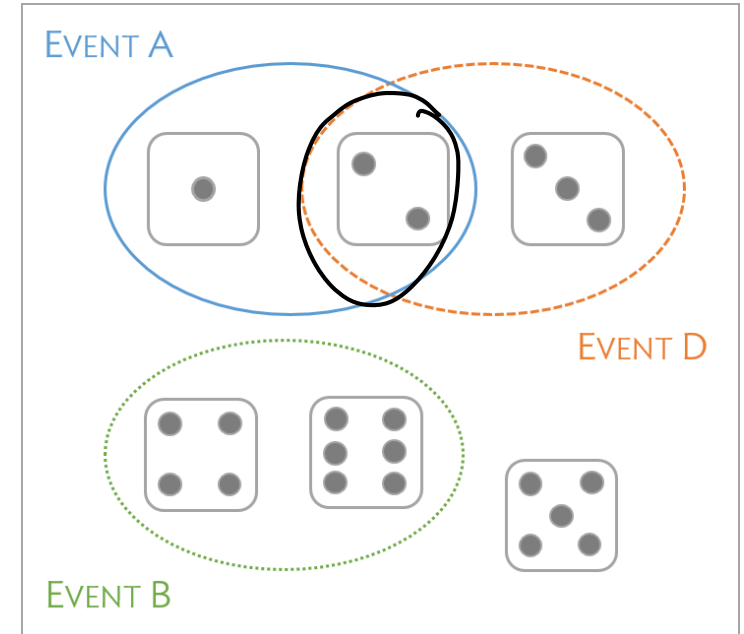
$$P(A \text{ or } D) = P(A) + P(D) - P(A \text{ and } D),$$

where $P(A \text{ and } D)$ is the probability that both events occur.



Think back to our die

- Event A and D are not disjoint, they share an outcome of rolling a 2
- How do we find the probability of event A or event D?



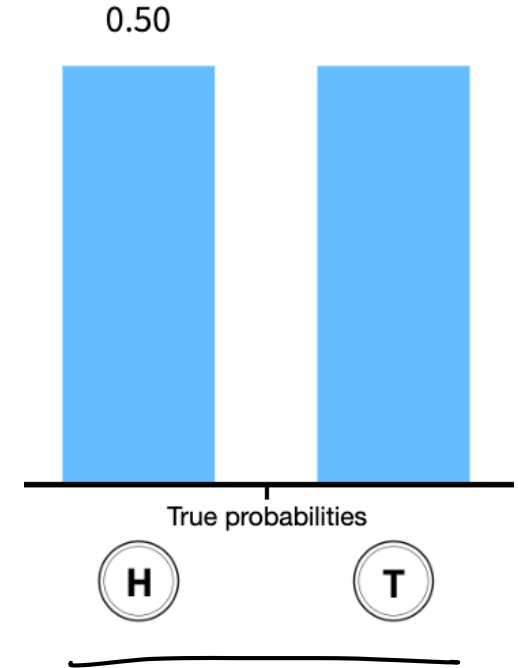
Probability distributions

- A **probability distribution** consists of all disjoint outcomes and their associated probabilities
- We've already seen one in our heads and tails example

Rules for a probability distribution

A probability distribution is a list of all possible outcomes and their associated probabilities that satisfies three rules:

1. The outcomes listed must be disjoint
2. Each probability must be between 0 and 1
3. The probabilities must total to 1



Complement of an event

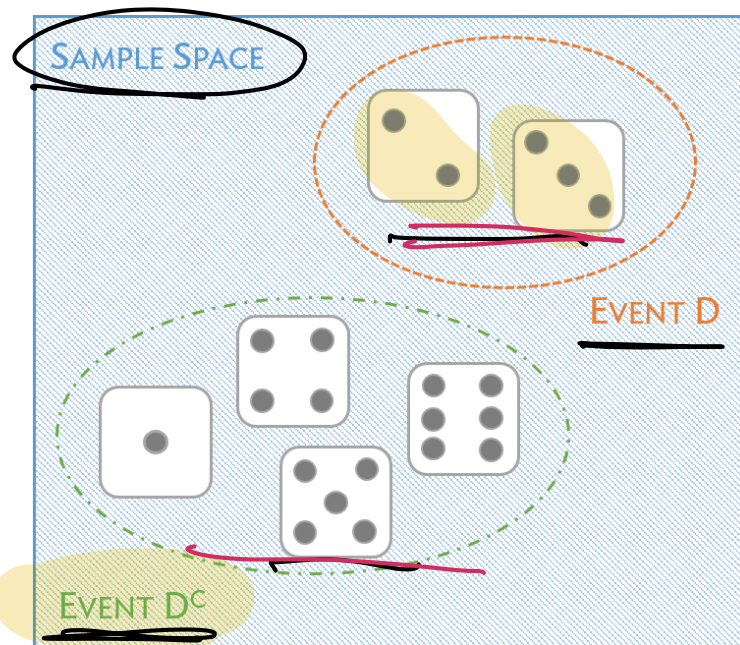
We need two math definitions for this:

- **Sample space:** denoted as S is the set of all possible outcomes
- **Complement:** complement of an event, say D , represents all the outcomes in the sample space that are not in D
 - Complement is denoted as D^c or D'

Complement

The complement of event A is denoted A^c , or A' , and A^c represents all outcomes not in A . A and A^c are mathematically related:

$$\underline{P(A) + P(A^c) = 1}, \text{ i.e. } \underline{P(A) = 1 - P(A^c)}.$$



$$P(D) + P(D')$$
$$\frac{2}{6} + \frac{4}{6} = 1$$

Independence

need 2 or more random phenomenon

- Two processes are **independent** if knowing the outcome of one provides no information about the outcome of the other
- For example, if we flip two different coins and one lands on heads, what does that tell us about the other coin?

Multiplication Rule for independent processes

If A and B represent events from two different and independent processes, then the probability that both A and B occur is given by:

$$P(A \text{ and } B) = P(A)P(B).$$

Similarly, if there are k events $A_1, \dots, A_k$ from k independent processes, then the probability they all occur is

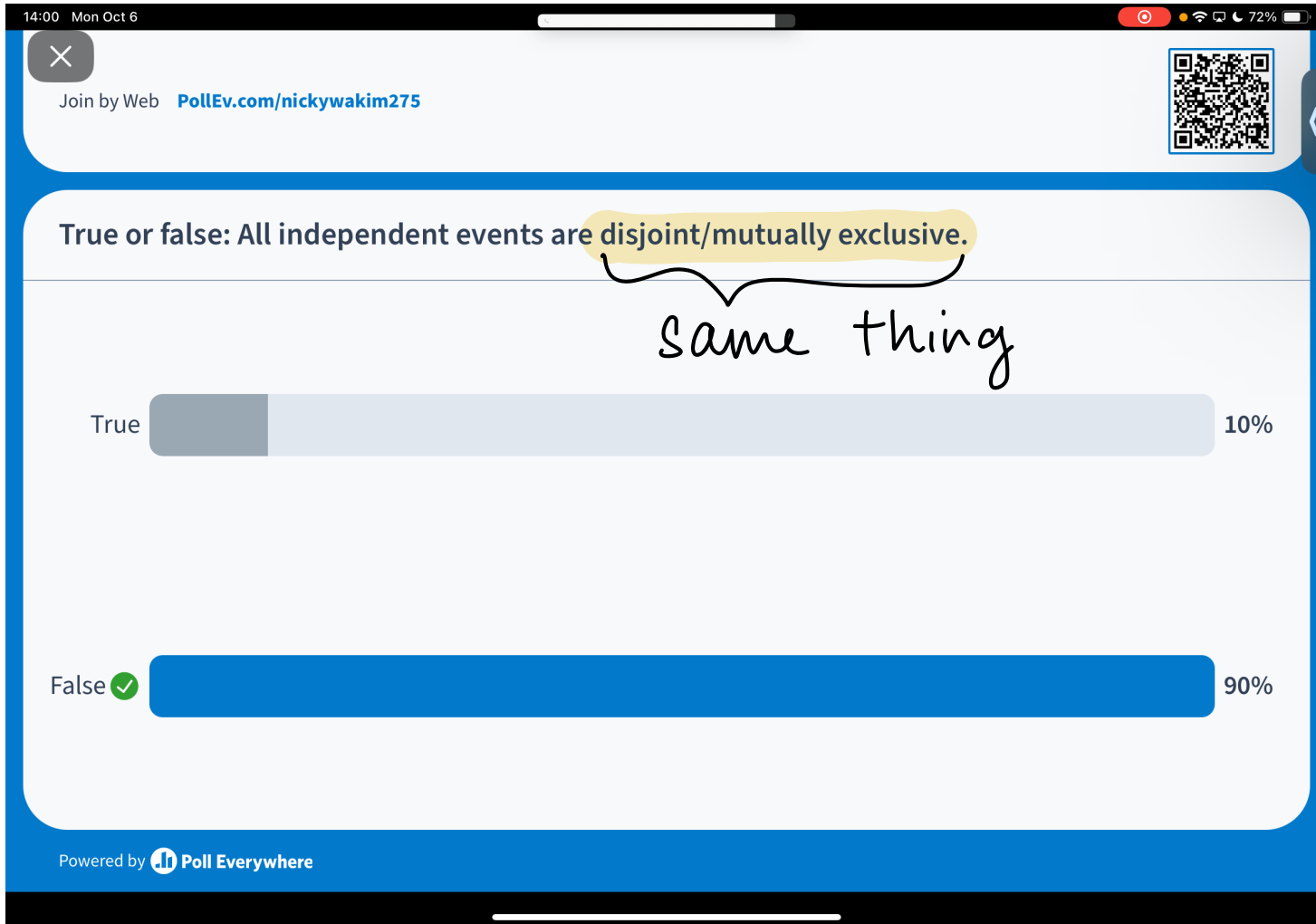
$$P(A_1)P(A_2) \cdots P(A_k).$$

A is flip 1 is a head $P(A) = 0.5$

B is flip 2 is a head $P(B) = 0.5$

$$\begin{aligned} P(A \text{ and } B) &= P(A)P(B) \\ &= 0.5 \cdot 0.5 \\ &= 0.25 \end{aligned}$$

Poll Everywhere Question 3



disjoint

$$P(A \text{ and } B) = 0$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

b/c = 0

independence

$$P(A \text{ and } B) = P(A) \cdot P(B) \neq 0$$

Learning Objectives

1. Define probability and explain the Law of Large Numbers within examples
2. Define relationships between events and their probability properties (including disjoint events, non-disjoint events, complements, and independent events)
3. Calculate an unknown probability in a word problem using the probability properties

Example rolling two dice

What is the probability that both dice will be 1?

$$\begin{aligned}
 &P(\text{blue is 1 \& green is 1}) \\
 &= P(\text{blue is 1}) P(\text{green is 1}) \\
 &= \left(\frac{1}{6}\right) \cdot \left(\frac{1}{6}\right) = \frac{1}{36}
 \end{aligned}$$

b/c independent

OUTCOMES OF THE BLUE DIE

	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

OUTCOMES OF THE GREEN DIE

other ex :

blue ≥ 4

blue & green sum to ≥ 7

$$P(\text{blue} \geq 4 \& \text{Blue} + \text{green} \geq 7) \neq P(\text{blue} \geq 4) \cdot P(\text{Blue} + \text{green} \geq 7)$$

NOT independent

36 poss outcomes

General steps for probability word problems

1. Define the events in the problem and make a Venn Diagram / viz
- ② Translate the words and numbers into probability statements
- ③ Translate the question into a probability statement
4. Think about the various definitions and rules of probabilities. Is there a way to define our question's probability statement (in step 3) using the probability statements with assigned values (in step 2)?
5. Plug in the given numbers to calculate the answer!

★ assume NOT disjoint
& NOT independent

Weekly medications

Example 3

If a subject has an

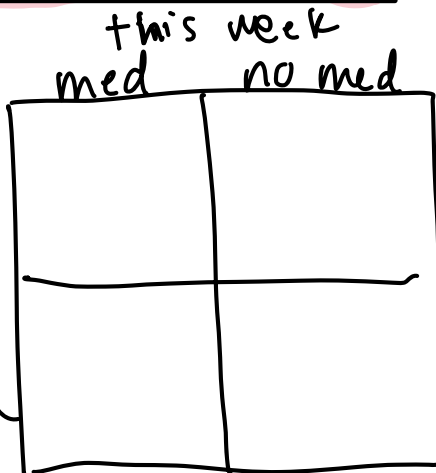
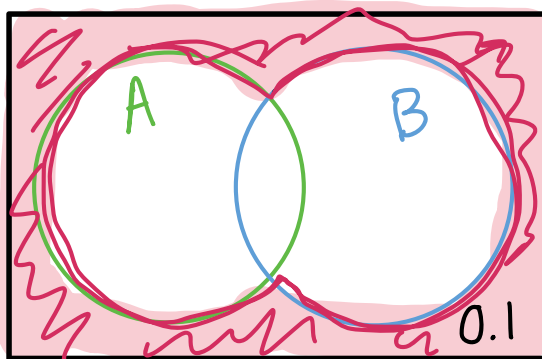
- 80% chance of taking their medication *this week*,
- 70% chance of taking their medication *next week*, and
- 10% chance of not taking their medication either week,

then find the probability of them taking their medication both weeks.

$U = \underline{A \text{ or } B \text{ or both}}$ *next week*

$\cap = A \text{ and } B$

Hint: Draw a Venn diagram labelling each of the parts to find the probability.



let A = take med this week

B = take med next week

$$P(A) = 0.8$$

$$P(B) = 0.7$$

$$P((A \cup B)^c) = 0.1$$

$$\text{let } F = A \cup B$$

$$P(F^c) = 0.1$$

$$P(F) + P(F^c) = 1$$

$$P(F) = 1 - P(F^c) = 1 - 0.1 = 0.9$$

↘ $A \cup B$

$$P(A \cup B) = 0.9$$

not disjoint

$$\begin{array}{l} \rightarrow \left[\begin{array}{l} P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) \\ - P(A \text{ or } B) \\ + P(A \text{ and } B) \end{array} \right. \left. \begin{array}{l} - P(A \text{ or } B) \\ + P(A \text{ and } B) \end{array} \right. \end{array}$$

Want to calc this

$$\begin{aligned} P(A \text{ and } B) &= P(A) + P(B) - P(A \text{ or } B) \\ &= 0.8 + 0.7 - 0.9 \\ &= 0.6 \end{aligned}$$

The probability of taking meds both weeks is 0.6.

if ind $P(A \text{ and } B) = P(A)P(B)$

$$0.6 \stackrel{?}{=} 0.8 \cdot 0.7$$

$$\neq 0.56$$

$\Rightarrow$ "thus" not independent